

Topic Models and Word Embedding Transcript

Now we are coming now to the actual textual part, analysing the textual part. Here we are trying to make use of the algorithms which try to use the semantic analysis and say group the contents to show us what is the kind of the email that they are discussing or what are the significant topics that are there in the email. So the relationship hypothesis is that if a word is being used in similar linguistic context as another like for example bankruptcy or credit default or stress in the finance.

These things are used in a similar semantic context signalling there is stress in the company. This can be thought of belonging to one particular topic and this is called word embedding. You are trying to group the words into particular categories depending upon their similarity of use in the text.

So once this is done, so a common example here would be to map the word king with man because a king is expected to be a man, so P is expected to be a woman. So you use P and you use woman, the algorithm would pick and put them together. The same way the text content in the emails has been analysed and you have found four or five different types of four different types of word clusters where you can completely place them belonging to same category.

For example, the court proceedings are in one particular area. Then you have the internal communications are in one communication area. Then you have the destroy, transmit, distribute, these are actions that are may be given.

Then you have the outcome penalty, protocol, comply, regulatory part in one area. So the algorithm is able to distinguish the contents and show that these are the types of the words that are there in particular set of emails. So maybe this can be categorised or looked as this discussing a particular topic.

So this is what we call as word embedding where you try to use the semantic similarity between the words and group them into topics. Then we have topic modelling. Here again it is an unsupervised machine learning technique.

There are these two algorithms LDA and LSA, we are not getting into the details. What they try to do is like they try to group the contents of a document or the documents into three or four topics. The bubble chart shows the inter topic distance.

Topic 1 is close to topic 2, but it is distant to topic 3. Likewise, it is distant to topic 4. So we can see the inter bubble distance as the deviation between the topics. We can say they are dissimilar topics, if they are closer they are similar topics. Then you have on the right hand side the frequency with which the words are there in the particular message or particular email.

So if you see the third, it is containing the words like these are the words and how much is the frequency of those words in those emails categorised as the topic. So topic model again is an algorithm, once you supply the data it would tell you like if you need three topics or four topics, it will distribute the space and show you. Once you supply data to the topic modelling, it is using either LDA algorithm or LSA algorithm, it will divide the emails and provide you three or four topics and also give you the distance similarities between these topics along with the frequency of the words.

We will be now looking at TNAC clustering, it is an enhanced textual algorithm helping you visualise and understand the topics that are there in the text. So again the logic here is it uses dimensionality reduction because textual data can be thought of having multiple dimensions higher dimensions, it will reduce the dimensions and create or mimic the distribution that is there in the original data. In this process, it is trying to reduce the computational load and try to efficiently present the topics that are there in the text.

So here also it will divide the documents or the rectangles are the documents and it will place the topics that are discussed in the documents as groups. You can see all these documents in the rectangles are likely to be orange topics which are discussed might be regarding bankruptcy, green might be action, blue might be fallout, red might be regulatory. So the topic in that particular document is dominated by discussion around this particular area.

So when we use TNAC clustering which stands for T-distributed stochastic neighbourhood embedding, so you are trying to divide the documents or the textual content into different topical regions so that you can vividly see which topics are most reflected in what kind of documents. So again either word embedding or your topic modelling or TNAC clustering are achieving only one objective. They are trying to divide or showcase you the topics dominant more than topics the dominant themes discussion areas that are there in the emails or in the textual document.

So you can use them to understand directly what is the text talking about or the different angles the text is talking about and then draw a conclusion from that. Like in dealing with structured data for unstructured data also you will have a very strong experiential learning session to strengthen these topics and get a look and feel of the algorithms. Thank you.

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